

Superoptimal Block-Circulant Preconditioner for Quasi-Block-Toeplitz Systems

1 Problem setup

Let $m \in \mathbb{N}$ and define $n = 2m + 1$. Consider the quasi-block-Toeplitz matrix $T \in \mathbb{C}^{nN \times nN}$ with block indices $k, j \in \{-m, \dots, m\}$:

$$T_{k,j} = (s + ik\omega)M \delta_{k,j} - A_{k-j}, \quad (1)$$

where:

- $M \in \mathbb{C}^{N \times N}$,
- $A_\ell \in \mathbb{C}^{N \times N}$ for $|\ell| \leq m$,
- $s \in \mathbb{C}$ and $\omega \in \mathbb{R}$.

We decompose:

$$T = D + S, \quad (2)$$

with

$$D_{k,j} = (s + ik\omega)M \delta_{k,j}, \quad (3)$$

$$S_{k,j} = -A_{k-j}. \quad (4)$$

2 Block-circulant preconditioner

Let $F \in \mathbb{C}^{n \times n}$ be the unitary DFT matrix:

$$F_{pq} = \frac{1}{\sqrt{n}} e^{-2\pi i pq/n}. \quad (5)$$

A block-circulant matrix C admits the diagonalization:

$$C = (F \otimes I_N) \Lambda (F^* \otimes I_N), \quad (6)$$

where $\Lambda = \text{blkdiag}(\Lambda_k)$.

We seek:

$$\min_{C \text{ block-circ.}} \|I - C^{-1}T\|_F^2. \quad (7)$$

3 Fourier-domain formulation

Define:

$$G = (F^* \otimes I)T(F \otimes I). \quad (8)$$

Using unitary invariance:

$$\|I - C^{-1}T\|_F^2 = \|I - \Lambda^{-1}G\|_F^2. \quad (9)$$

Expanding in block form:

$$= \sum_{k,j} \|\delta_{k,j}I - \Lambda_k^{-1}G_{k,j}\|_F^2. \quad (10)$$

Separate diagonal and off-diagonal terms:

$$= \sum_k \|I - \Lambda_k^{-1}G_{k,k}\|_F^2 + \sum_k \sum_{j \neq k} \|\Lambda_k^{-1}G_{k,j}\|_F^2. \quad (11)$$

Thus the minimization decouples over k .

4 Optimality condition

Let $X_k = \Lambda_k^{-1}$. Define:

$$f(X_k) = \|I - X_k G_{k,k}\|_F^2 + \sum_{j \neq k} \|X_k G_{k,j}\|_F^2. \quad (12)$$

Expanding:

$$f(X_k) = N - \text{tr}(X_k G_{k,k}) - \text{tr}(G_{k,k}^* X_k^*) + \text{tr}(X_k R_k X_k^*), \quad (13)$$

where:

$$R_k = \sum_j G_{k,j} G_{k,j}^*. \quad (14)$$

Differentiating with respect to X_k^* :

$$\frac{\partial f}{\partial X_k^*} = -G_{k,k} + X_k R_k = 0. \quad (15)$$

Hence:

$$X_k = G_{k,k} R_k^{-1}. \quad (16)$$

Therefore:

$$\boxed{\Lambda_k^{\text{opt}} = R_k G_{k,k}^{-1}} \quad (17)$$

5 Explicit expression for $G_{k,k}$

Using the structure of T :

$$G_{k,k} = (f_k^* \otimes I)T(f_k \otimes I), \quad (18)$$

where f_k is the k -th Fourier vector.

The diagonal part gives:

$$(f_k^* D f_k) = (s + ik\omega)M. \quad (19)$$

The Toeplitz part gives:

$$(f_k^* S f_k) = \sum_{\ell=-m}^m A_\ell e^{ik\ell\theta}, \quad (20)$$

with $\theta = 2\pi/n$.

Thus:

$$G_{k,k} = (s + ik\omega)M - \sum_{\ell=-m}^m A_\ell e^{ik\ell\theta}. \quad (21)$$

6 Expression for R_k via Parseval

We use:

$$R_k = \sum_j G_{k,j} G_{k,j}^* = (f_k^* \otimes I) T T^* (f_k \otimes I). \quad (22)$$

Expanding:

$$R_k = \frac{1}{n} \sum_{p,q} e^{ik(q-p)\theta} (T T^*)_{p,q}. \quad (23)$$

Group by $m = q - p$:

$$R_k = \sum_{m=-2m}^{2m} B_m e^{ikm\theta}, \quad (24)$$

where:

$$B_m = \frac{1}{n} \sum_p (T T^*)_{p,p+m}. \quad (25)$$

7 Structure of TT^*

We have:

$$TT^* = DD^* + DS^* + SD^* + SS^*. \quad (26)$$

- SS^* is block Toeplitz (convolution of A_ℓ)
- DD^* and mixed terms depend on k

Thus TT^* is block-banded but not Toeplitz.

8 Efficient computation

8.1 Computation of $G_{k,k}$

For all k , compute:

$$G_{k,k} = (s + ik\omega)M - \sum_{\ell=-m}^m A_\ell e^{ik\ell\theta}. \quad (27)$$

This is a matrix-valued DFT:

- Assemble $\{A_\ell\}$ into a sequence
- Apply FFT to each matrix entry

Cost:

$$\mathcal{O}(n \log n \cdot \text{nnz per block})$$

8.2 Computation of R_k

Steps:

1. Compute TT^* exploiting sparsity (bandwidth $2m$)
2. Extract diagonal averages:

$$B_m = \frac{1}{n} \sum_p (TT^*)_{p,p+m}$$

3. Compute FFT:

$$R_k = \sum_m B_m e^{ikm\theta}$$

Cost:

$$\mathcal{O}(nm \cdot \text{nnz}) + \mathcal{O}(n \log n \cdot N^2)$$

8.3 Application of C^{-1}

Given v :

1. FFT:

$$\hat{v} = (F \otimes I)v$$

2. For each k :

$$R_k y_k = \hat{v}_k \tag{28}$$

$$x_k = G_{k,k} y_k \tag{29}$$

3. Inverse FFT:

$$x = (F^* \otimes I)\hat{x}$$

Cost per iteration:

$$\mathcal{O}(n \log n) + n \cdot \mathcal{O}(N^3)$$

9 Interpretation

$$\Lambda_k^{\text{opt}} = G_{k,k} + \left(\sum_{j \neq k} G_{k,j} G_{k,j}^* \right) G_{k,k}^{-*}. \tag{30}$$

Thus the preconditioner corrects for off-diagonal energy in G .

10 Special case

If T is block-circulant:

$$G_{k,j} = 0 \ (k \neq j) \Rightarrow \Lambda_k = G_{k,k}$$